Lec1

Meiying Zhang

COE201 - AI Large Models

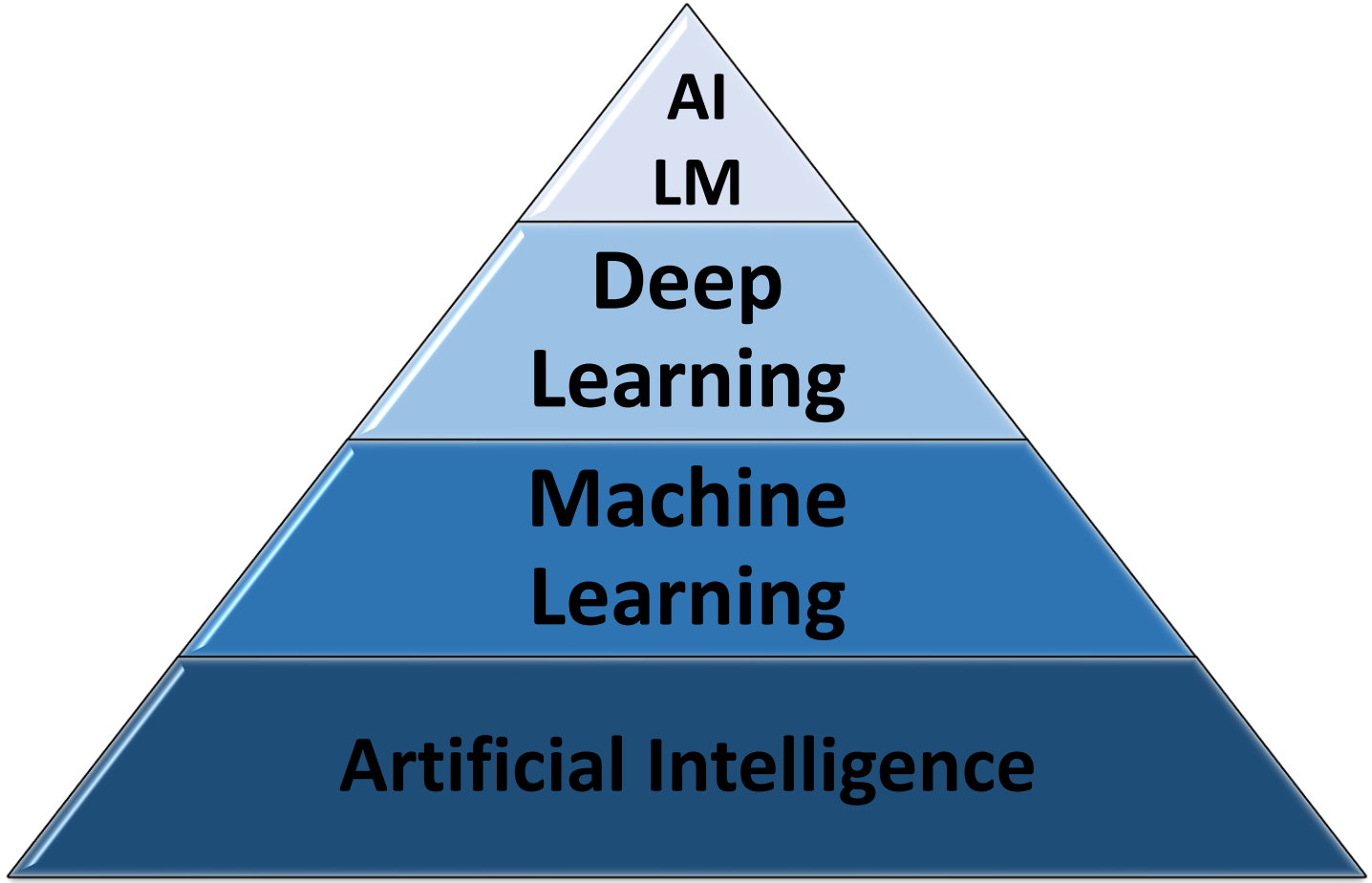
- Lectures: Monday 10:20-12:10AM
- Software labs: Monday 4:20-6:10PM
- Office hours: Thursday 2:30-3:30PM
- For personal /sensitive matters: zhangmy@sustech.edu.cn

Agenda

1. The evolution of AI Large Model
2. What is an AI Large Model?
3. The essence of AI Large Model

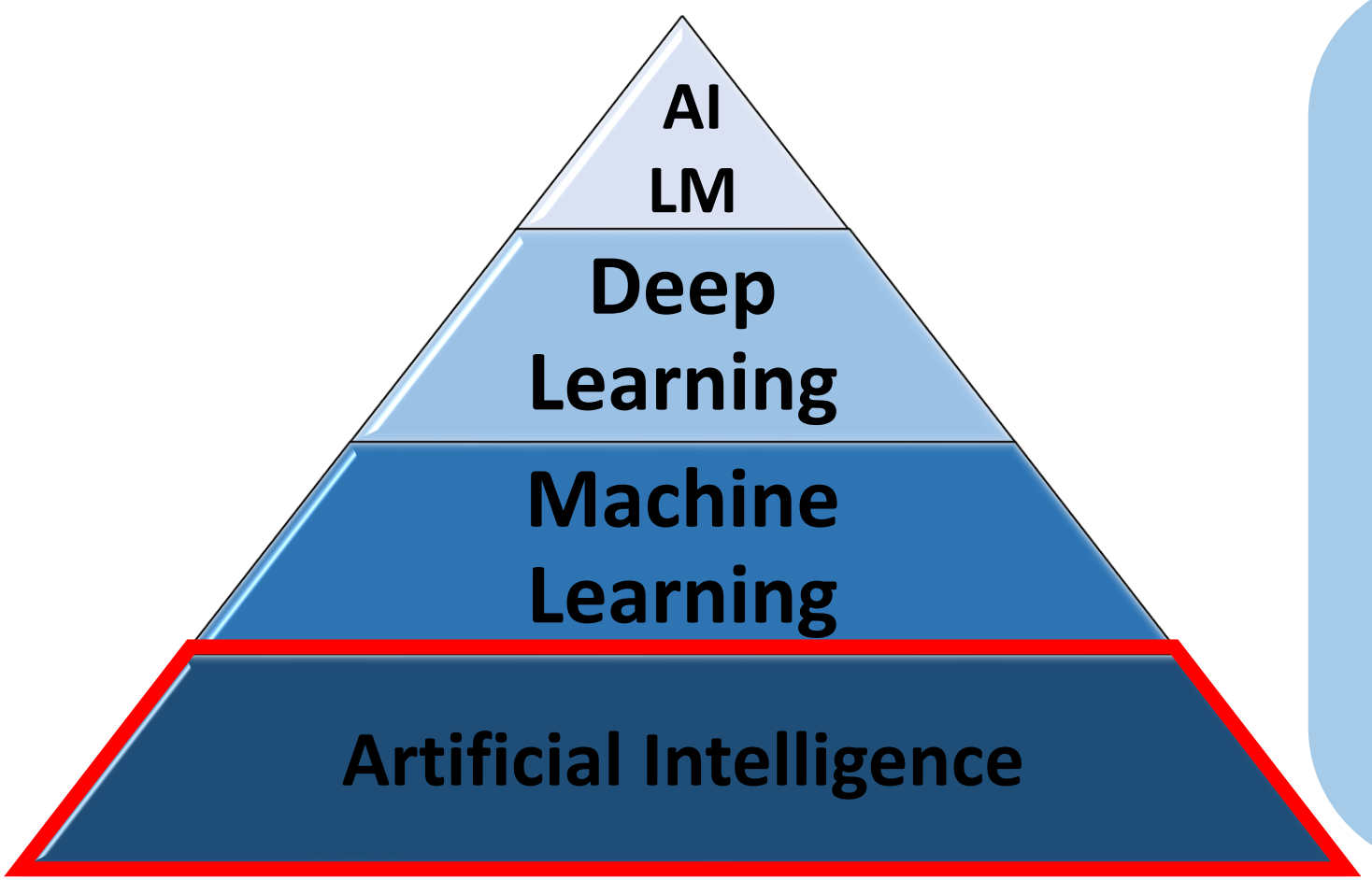
LM Progression

The evolution of AI Large Model?



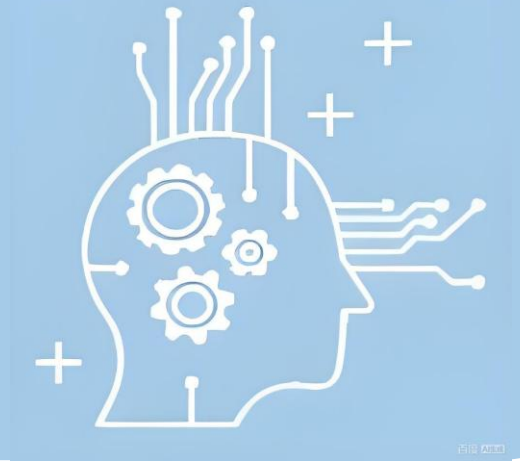
LM Progression

The evolution of AI Large Model?



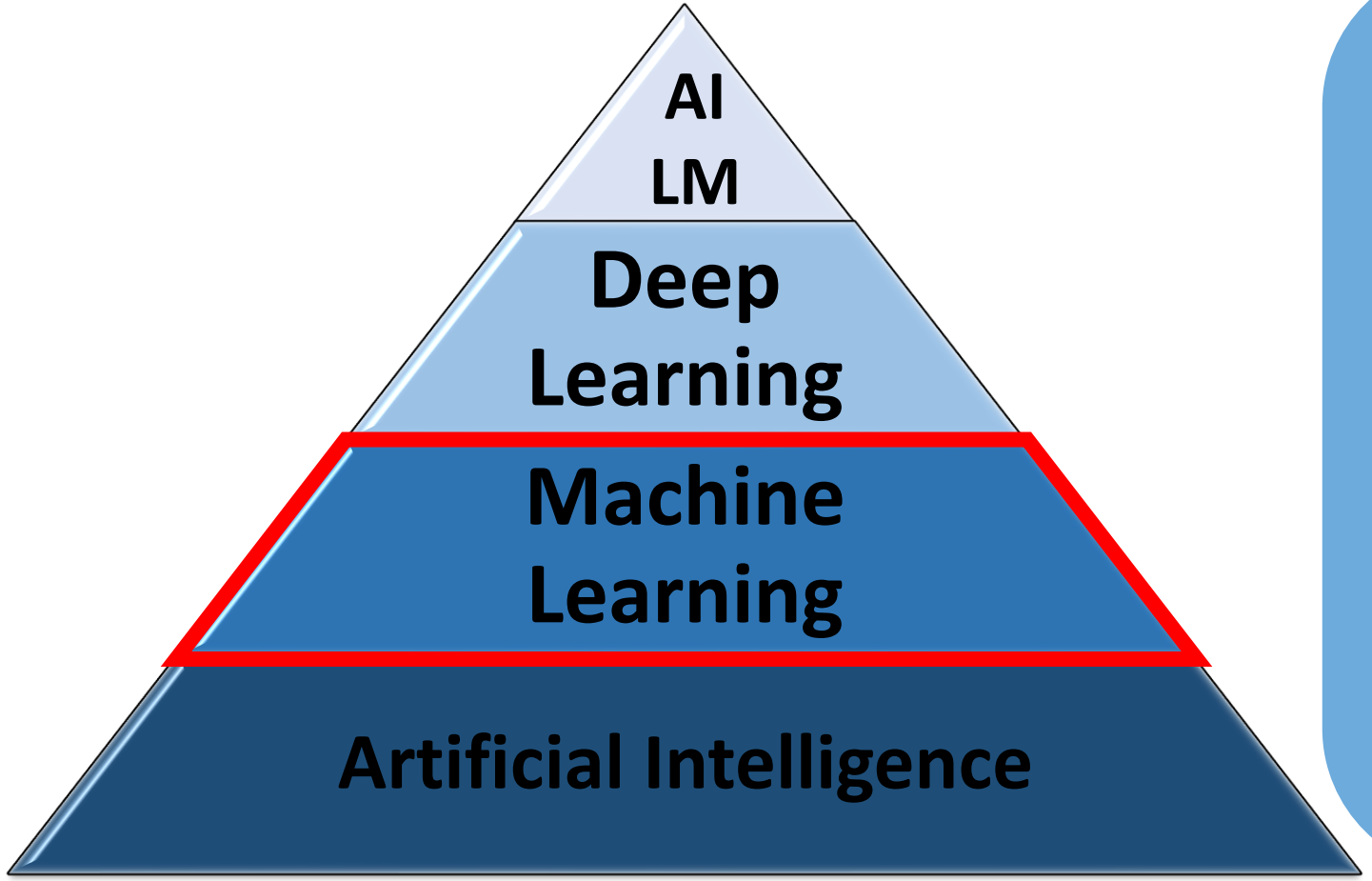
ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



LM Progression

The evolution of AI Large Model?

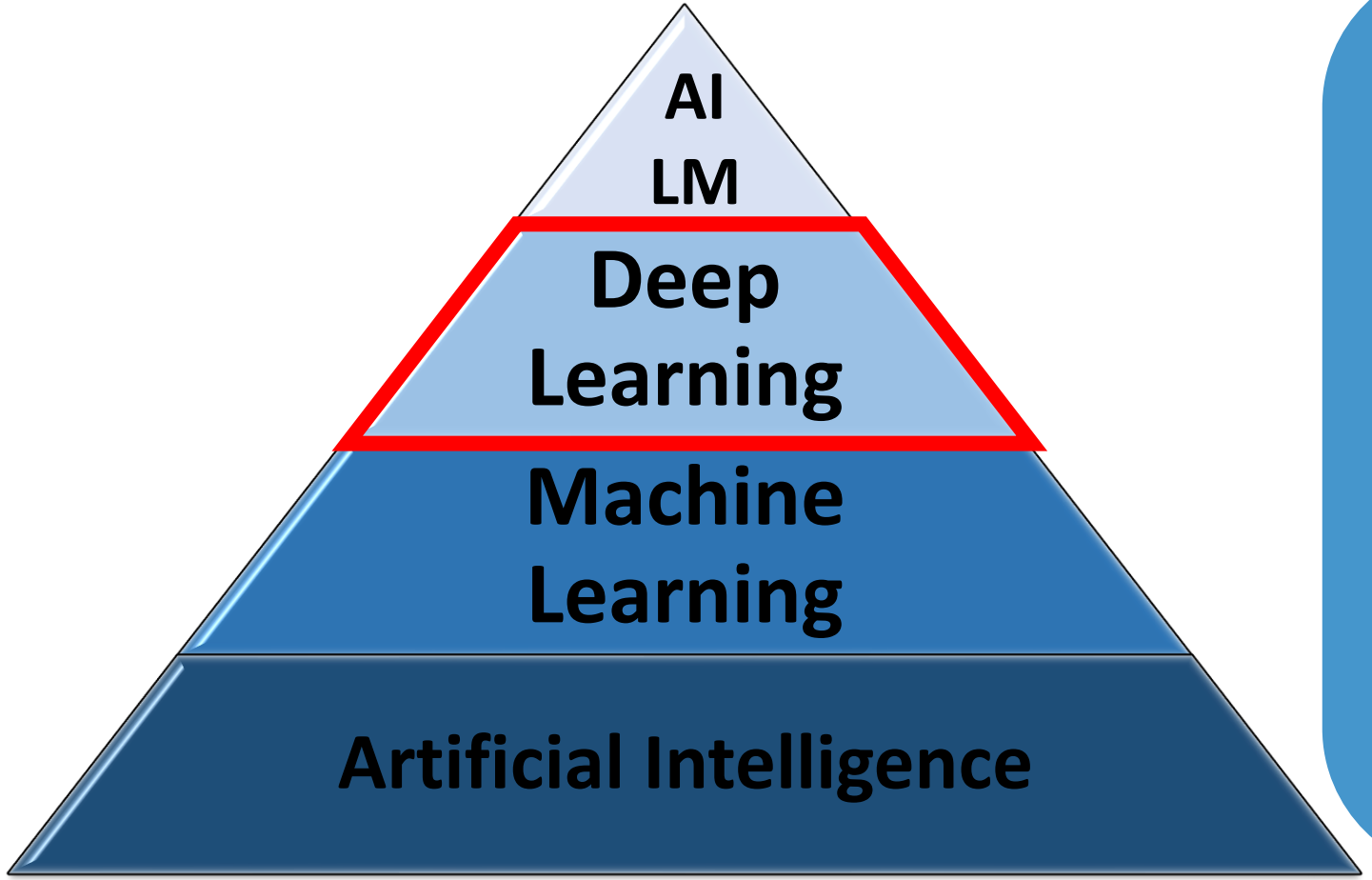


MACHINE LEARNING

Ability to learn without explicitly being programmed

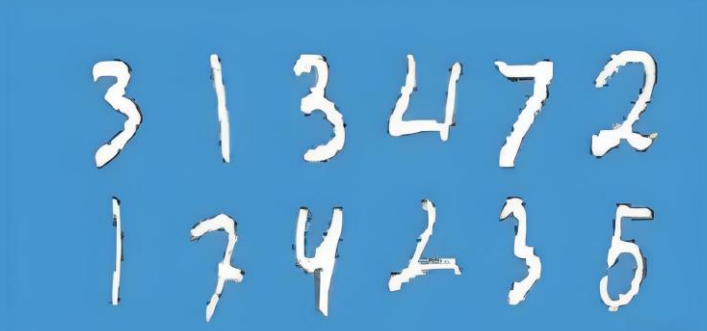
LM Progression

The evolution of AI Large Model?



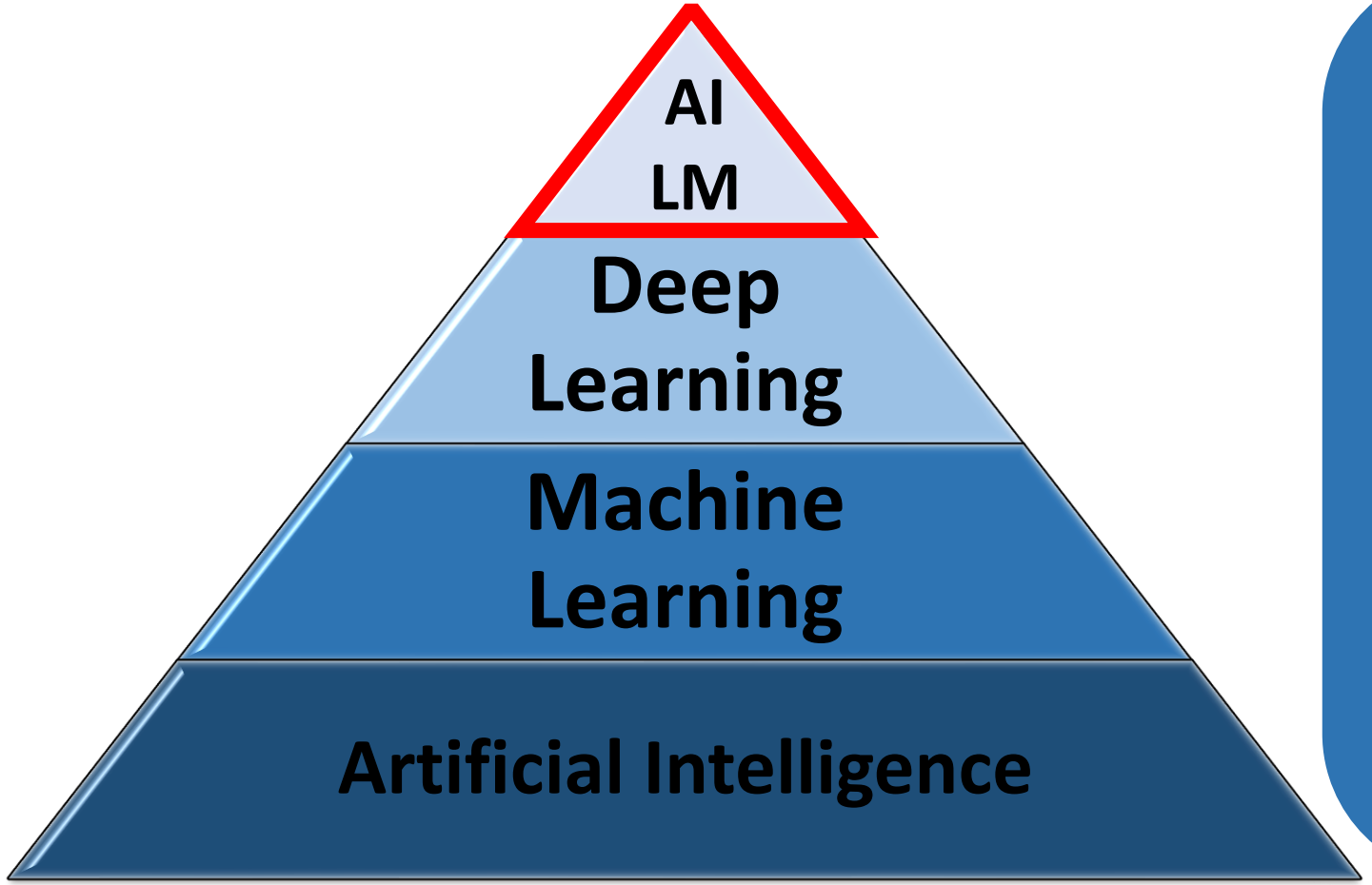
Deep Learning

Extract patterns from data using neural networks



LM Progression

The evolution of AI Large Model?

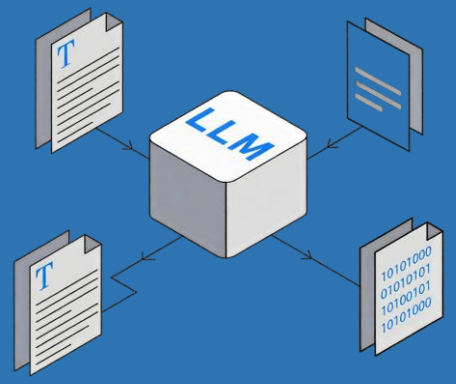


AI Large Model

That is "*Generative AI*"

focuses on creating new

content, e.g., text, video…



LM Progression

1952 — Stochastic Gradient Descent

1958 — Perceptron
- Learnable Weights

⋮

1986 — Backpropagation
- Multi-Layer Perceptron

1995 — Deep Convolutional NN
- Digit Recognition

⋮

2018 — Transformer
- GPT, BERT

⋮

Why LM dominate Now?

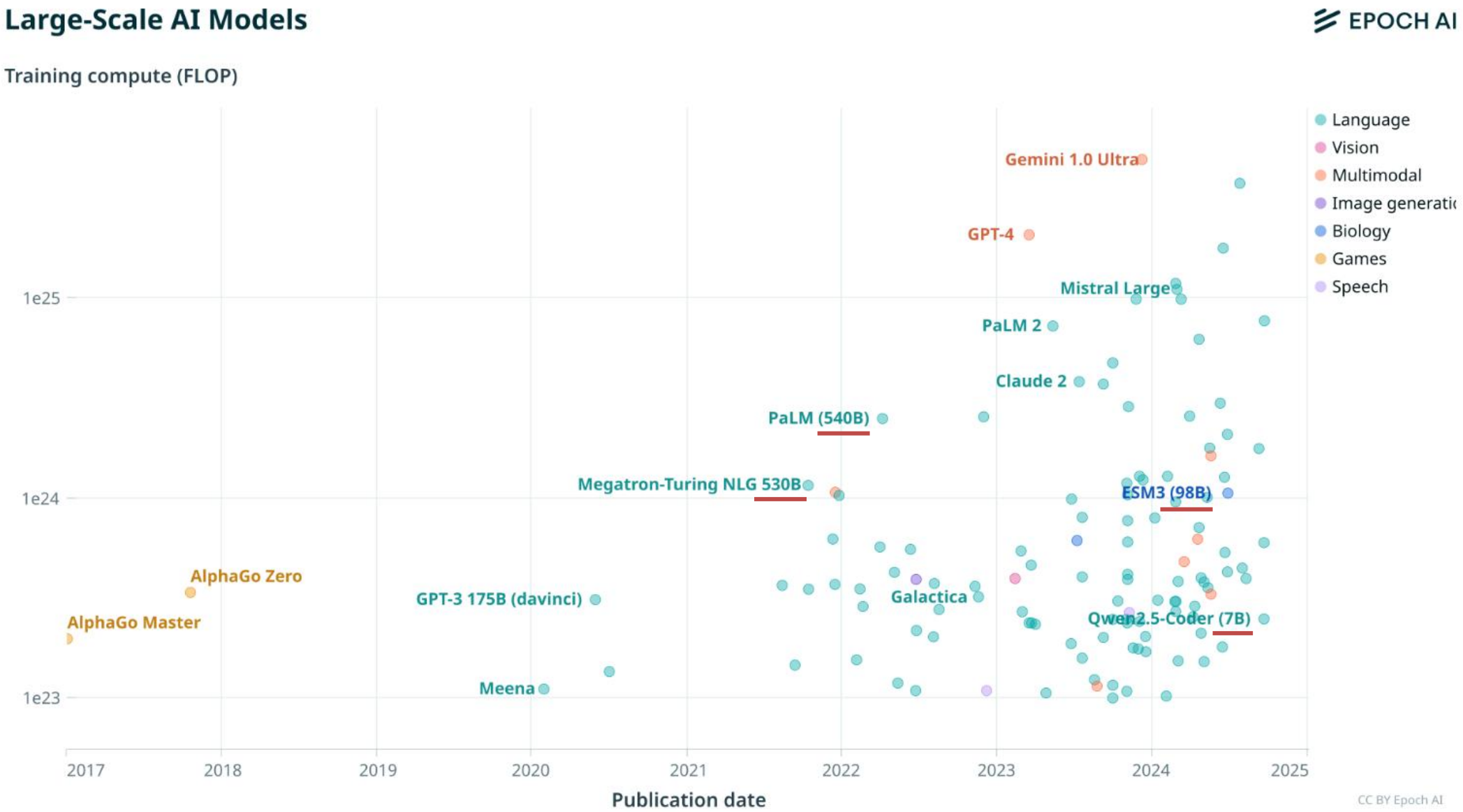
1. Big Data
- Larger Datasets
- Easier Collection & Storage

WIKIPEDIA
The Free Encyclopedia

2. Hardware
- Graphics Processing Units (GPUs)
- Massively Parallelizable

3. Software
- Improved Techniques
- New Models
- Toolboxes

LM Progression



What is an AI Large Model?

AI Large Model

Sora (OpenAI),Veo, Genie (Google)…are video generation LMs

A serene straight street lined with white houses on both sides and heavily covered in snow. The ground is blanketed with a thick layer of snow, and footprints are clearly visible with no pedestrians or passing vehicles. On both sides of the street, some green vegetation and bushes peek through the snow, appearing slightly wilted. The entire scene emits a sense of tranquility and coldness. However, due to the fog, the entire scene are blurred and indistinguishable, offering a glimpse into a peaceful urban neighborhood.

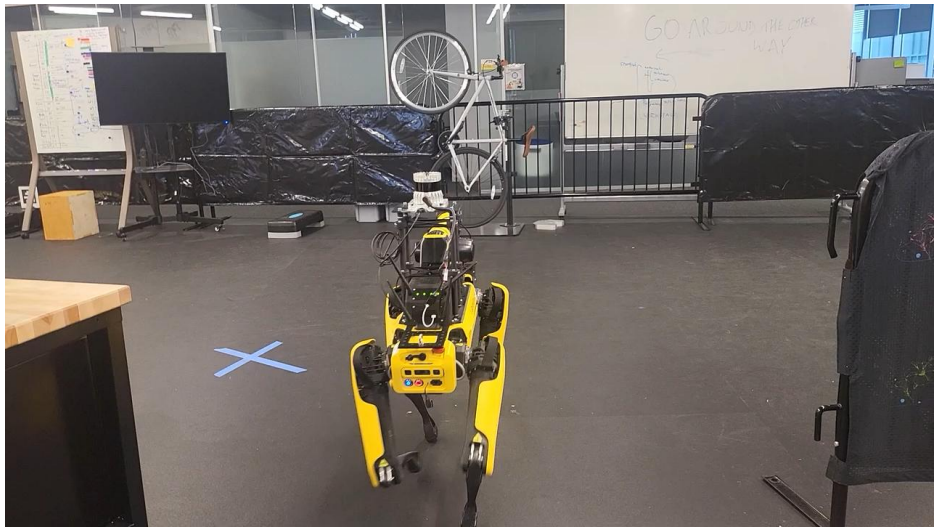
→ Large Model →

AI Large Model

RynnBrain, LingBot (Ali), Tairos …are video generation LMs

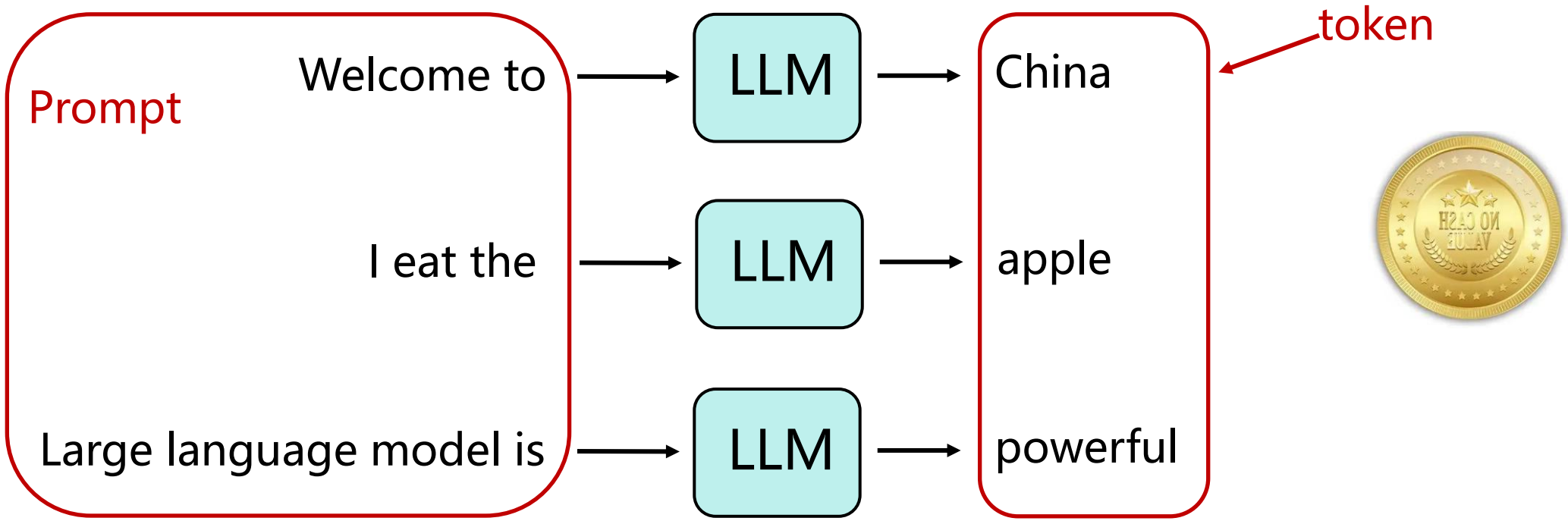
Move forward to the bicycle. Turn right, then move to the chair. Turn left and stop near the plant.

→ Large Model →



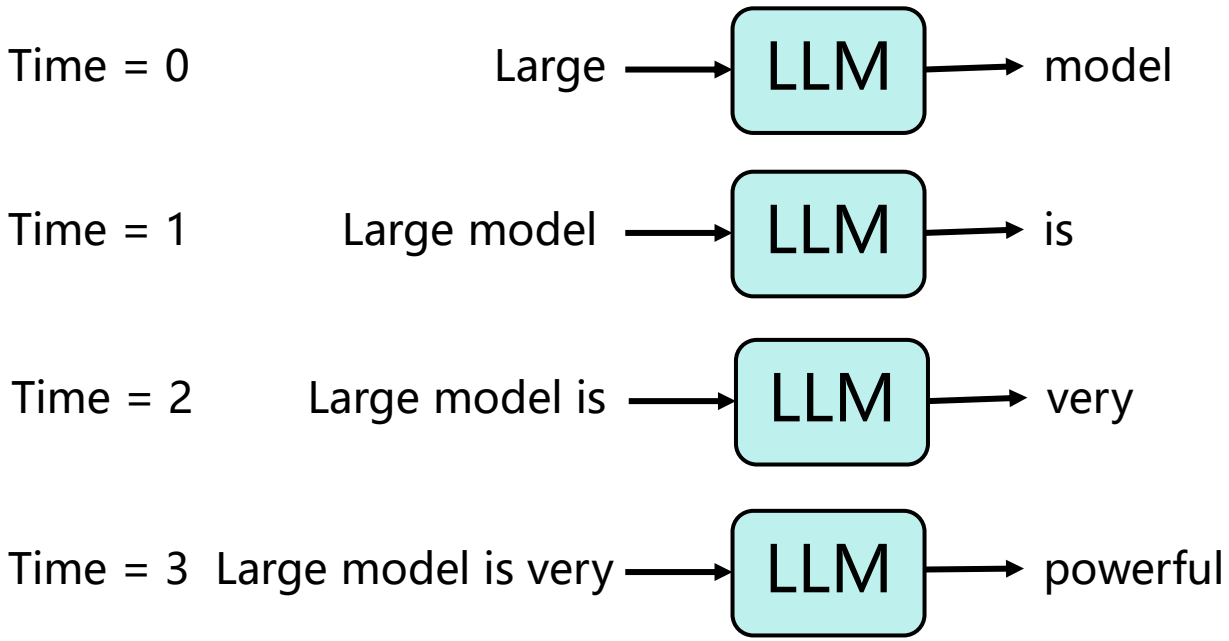
AI Large Model

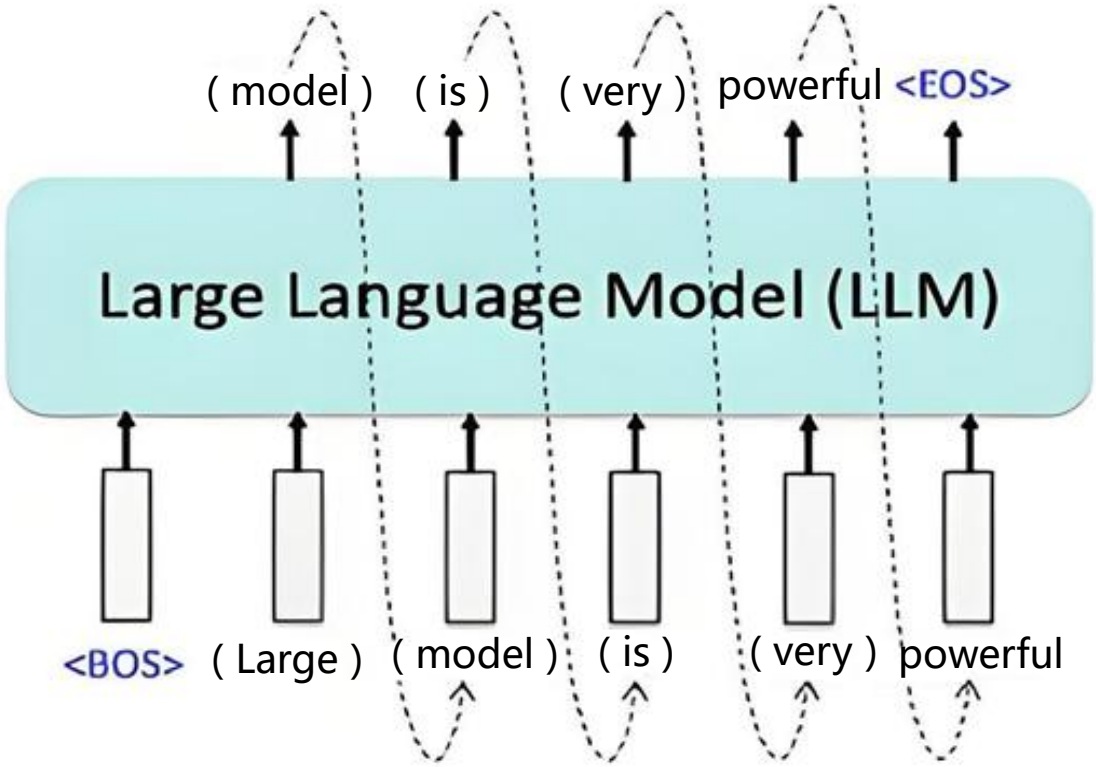
ChatGPT, Deepseek, Gemini, Claude…are Large Language Models (LLMs)



AI Large Model

How does a language model respond?





AI Large Model

How does the model know which one to follow?

Token distribution

I eat the → LLM →

Token	Score
apple	0.02
banana	0.25
ice	0.017
how	0.0000…
#	0.0000…
en	0.0000…
……	

Vocabulary

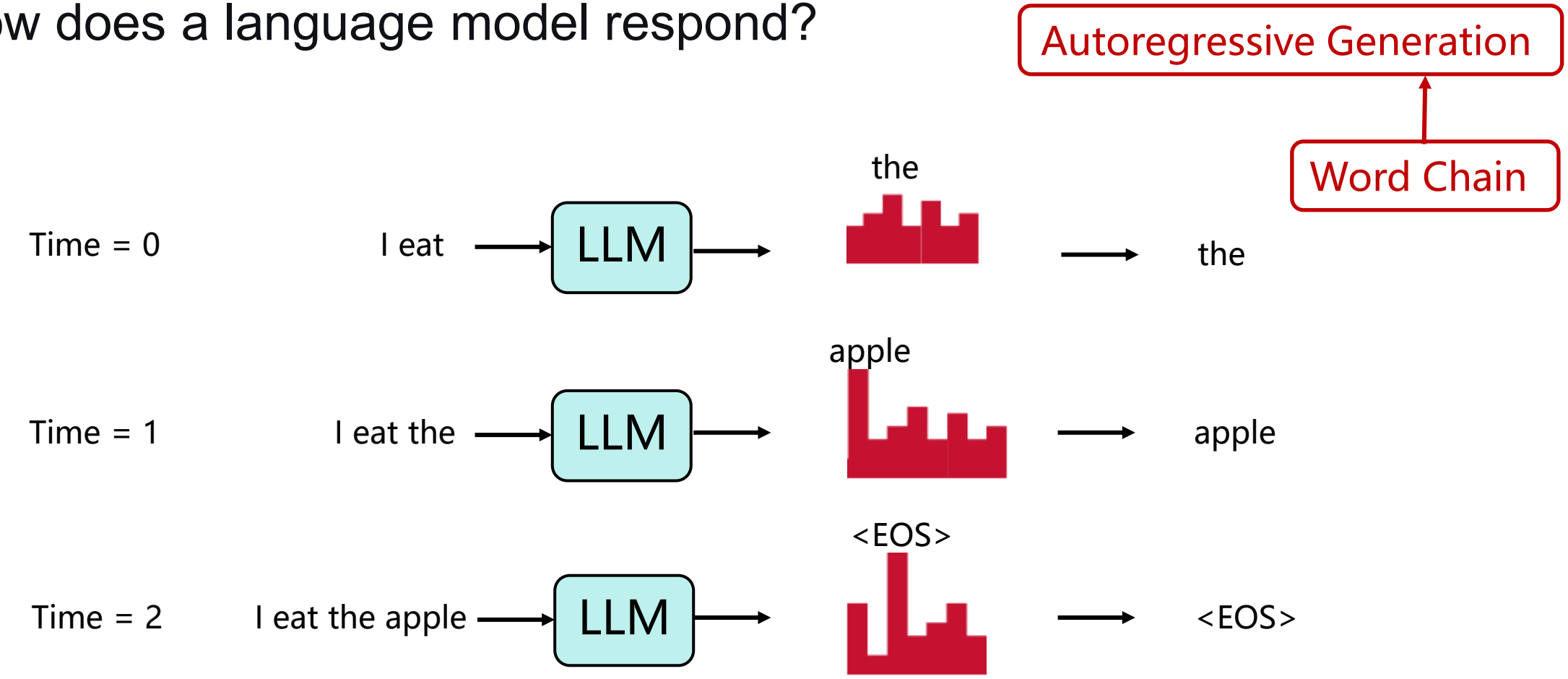
Probability

Sampling → apple

2%

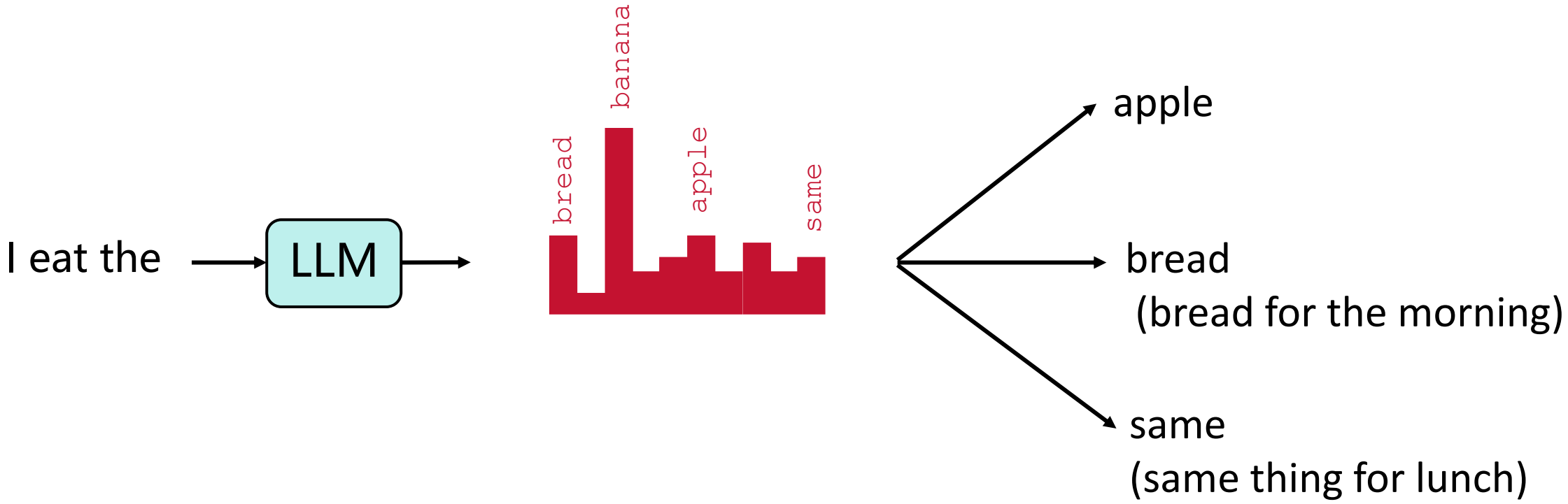
AI Large Model

How does a language model respond?



AI Large Model

Why is the answer different each time?



AI Large Model

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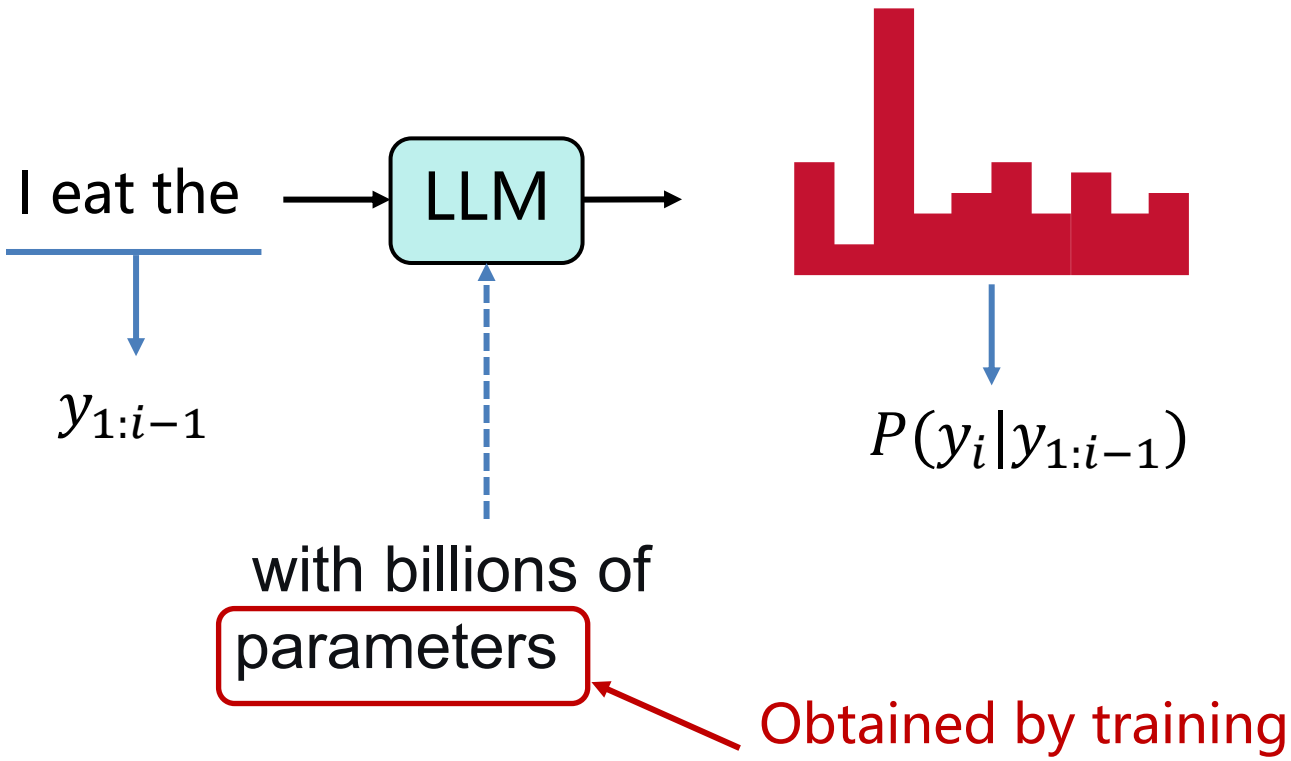


AI Large Model

Is the word chain easy?

Language Knowledge:

red ____ → LLM → noun (high probability)
verb (low probability)

World Knowledge:

The boiling point of water is __ → LLM → 100°C (high probability)
0°C (low probability)

At 0.5 ATM, the boiling point of water is __ → LLM → 100°C (low probability)

What is the essence of AI Large Model?

AI Large Model

A language model (LM) computes likelihood of a language sequence (e.g., words or tokens*).

$$P(y_1, y_2, \ldots, y_n)$$

$$\mathrm{P}(I, eat, the, apple)$$

Operationalized as outputting a probability distribution over the next token in a sequence given the previous tokens in the sequence.

$$P(y_i \mid y_{1:i-1})$$

$$\mathrm{P}(apple | I, eat, the)$$

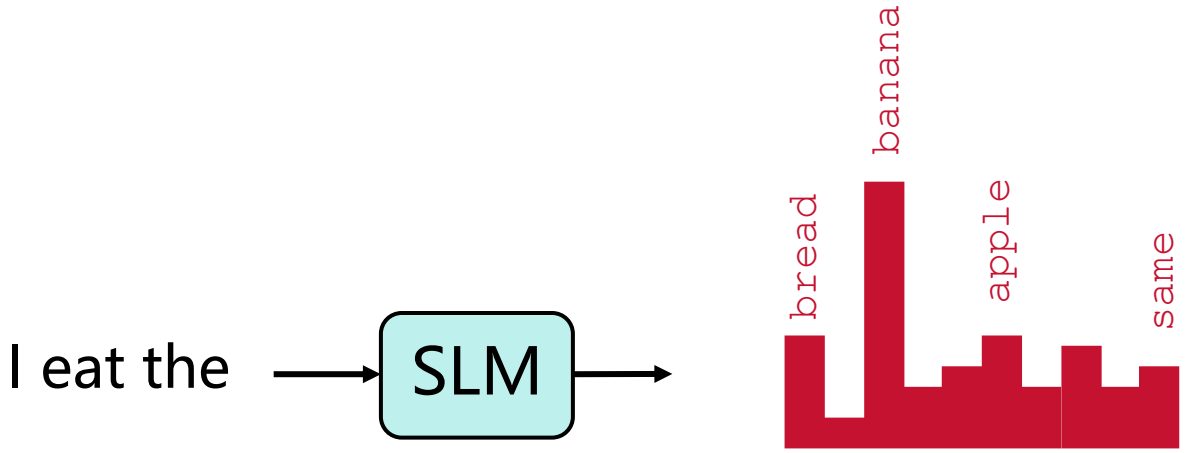
Historically, language models were statistical n-gram models.

AI Large Model

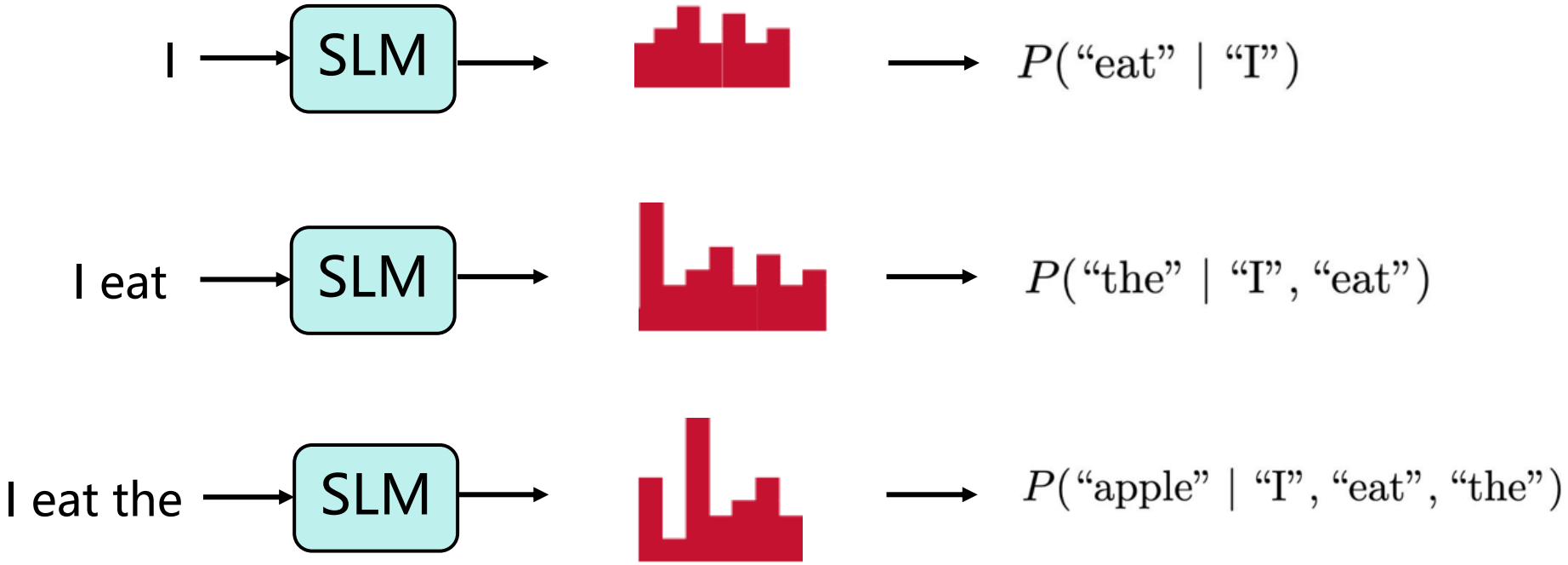
Suppose we are building a statistical language model (SLM) using a specific text corpus `C`.

We observe that the word "`apple`" follows the words "`I eat the`" 2% of the times that "`I eat the`" occurs in `C`.

$$P(\text{“apple”} \mid \text{“I”}, \text{“eat”}, \text{“the”}) = 0.02$$



AI Large Model



Computing Sequence Likelihood

Sequence likelihood can be computed from our LM using the chain rule:

$$P(\text{“I”}, \text{“eat”}, \text{“the”}, \text{“apple”}) = P(\text{“apple”} \mid \text{“I”}, \text{“eat”}, \text{“the”}) \times P(\text{“the”} \mid \text{“I”}, \text{“eat”}) \times P(\text{“eat”} \mid \text{“I”}) \times P(\text{“I”})$$

Chain Rule

More abstractly:

$$P(y_1, y_2, \ldots, y_n) = P(y_n \mid y_1, \ldots, y_{n-1}) \times \underbrace{P(y_{n-1} \mid y_1, \ldots, y_{n-2}) \times \ldots P(y_1)}_{P(y_1, y_2, \ldots, y_{n-1})}$$

N-gram versus Neural Models

How can we compute the conditional probability of the `i`-th token given the preceding tokens?

$$P(y_i \mid y_{1:i-1}) = \frac{N(y_{1:i})}{N(y_{1:i-1})}$$

N-gram language model approach:

$$P(y_i \mid y_{1:i-1}) \approx \frac{N(y_{i-n+1:i})}{N(y_{i-n+1:i-1})}$$

2-gram language model example

$$\boxed{\begin{aligned} &P(\text{“I”}, \text{“eat”}, \text{“the”}, \text{“apple”}) = \\ &\qquad P(\text{“apple”} \mid \cancel{\text{“I”}, \text{“eat”}}, \text{“the”}) \times \\ &\qquad\qquad P(\text{“the”} \mid \cancel{\text{“I”}}, \text{“eat”}) \times P(\text{“eat”} \mid \text{“I”}) \times P(\text{“I”}) \end{aligned}}$$

N-gram versus Neural Models

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3-gram language model example

$$\boxed{\begin{aligned} &P(\text{“I”}, \text{“eat”}, \text{“the”}, \text{“apple”}) = \\ &\qquad P(\text{“apple”} \mid \cancel{\text{“I”}}, \text{“eat”}, \text{“the”}) \times \\ &\qquad\qquad P(\text{“the”} \mid \text{“I”}, \text{“eat”}) \times P(\text{“eat”} \mid \text{“I”}) \times P(\text{“I”}) \end{aligned}}$$

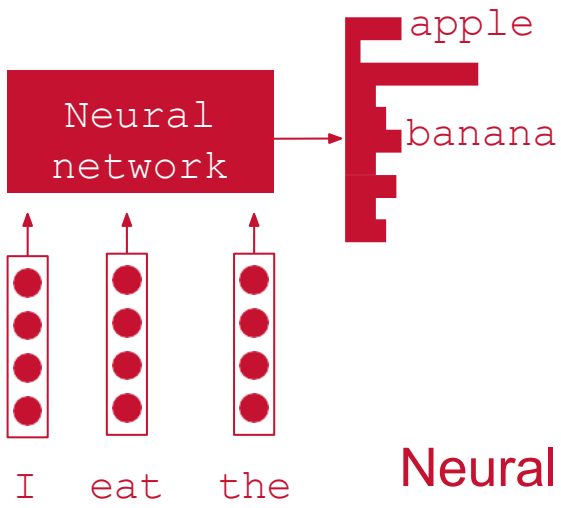
N-gram versus Neural Models

How can we compute the conditional probability of the `i`-th token given the preceding tokens?

$$P(y_i \mid y_{1:i-1}) = \frac{N(y_{1:i})}{N(y_{1:i-1})}$$

Neural language model approach:

$$P(y_i \mid y_{1:i-1}) \approx NN(y_{1:i-1})$$



Neural language model example

$$P(\text{“I”}, \text{“eat”}, \text{“the”}, \text{“apple”}) = \underline{NN}(\text{“apple”} \mid \text{“I”}, \text{“eat”}, \text{“the”}) \times \underline{NN}(\text{“the”} \mid \text{“I”}, \text{“eat”}) \times \underline{NN}(\text{“eat”} \mid \text{“I”}) \times \underline{NN}(\text{“I”})$$

Neural Language Models: Conditioned v. Unconditioned

Neural language models can be designed to …

…predict the next word given the previous ones.

- May be referred to as unconditioned.
  $P(Y) = P(y_1, y_2, \ldots, y_n)$
- At each step the LM predicts:
  $P(y_i \mid y_{1:i-1})$
- Examples:
  - GPT-*
  - LLaMA

…predict the next word given the previous ones *and* an additional conditioning sequence.

- May be referred to as conditioned.
  $P(Y \mid X) = P(y_1, y_2, \ldots, y_n \mid x_1, x_2, \ldots, x_m)$
- At each step the LM predicts:
  $P(y_i \mid y_{1:i-1}, x_{1:m})$
- Examples:
  - T5
  - machine translation models
- Sometimes called seq2seq models.

Neural Language Models: Conditioned v. Unconditioned

Theoretically, any task designed for an unconditional LM can be turned into one for a conditional LM, and vice-versa.

Task: Continue the sequence

Unconditioned

$$P(Y = [\text{"At"}, \text{"0.5"}, \text{"ATM"}, \text{"the"}, \text{"boiling"}, \text{"point"}, \text{"of"}, \text{"water"}, \text{"}is\text{"}])$$

$$P(y_i = \text{"point"} \mid Y_{1:i-1} = [\text{"At"}, \text{"0.5"}, \text{"ATM"}, \text{"the"}, \text{"boiling"}])$$

Conditioned

$$P(\text{Y}=[\text{"the"}, \text{"boiling"}, \text{"point"}, \text{"of"}, \text{"water"}, \text{"}is\text{"}], X = [\text{"At"}, \text{"0.5"}, \text{"ATM"}])$$

$$P(y_i = \text{"point"} \mid Y_{1:i-1}=[\text{"the"}, \text{"boiling"}], X = [\text{"At"}, \text{"0.5"}, \text{"ATM"}])$$

Neural Language Models: Conditioned v. Unconditioned

Task: Translate from English to French

Unconditioned

$$P(Y = [\text{“English”}, \text{“:”}, \text{“The”}, \text{“hippo”}, \text{“ate”}, \text{“my”}, \text{“homework”}, \text{“French”}, \text{“:”}, \text{“L’hippopotame”}, \text{“a”}, \text{“mangé”}, \text{“mes”}, \text{“devoirs”}])$$

$$P(y_i = \text{“mangé”} \mid Y_{1:i-1} = [\text{“English”}, \text{“:”}, \text{“The”}, \text{“hippo”}, \text{“ate”}, \text{“my”}, \text{“homework”}, \text{“French”}, \text{“:”}, \text{“L’hippopotame”}, \text{“a”}])$$

Conditioned

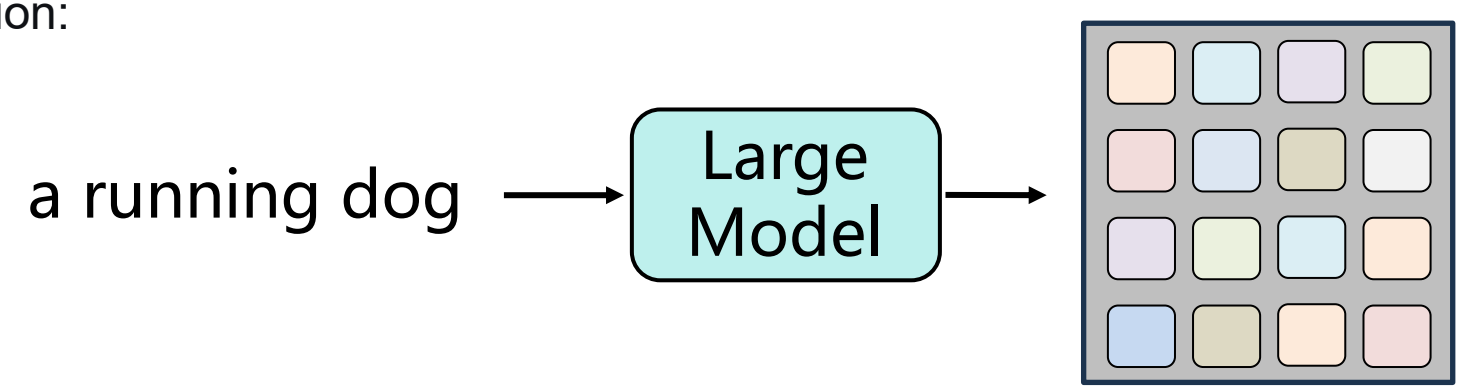
$$P(Y = [\text{“L’hippopotame”}, \text{“a”}, \text{“mangé”}, \text{“mes”}, \text{“devoirs”}], X = [\text{“The”}, \text{“hippo”}, \text{“ate”}, \text{“my”}, \text{“homework”}])$$

$$P(y_i = \text{“mangé”} \mid Y_{1:i-1} = [\text{“L’hippopotame”}, \text{“a”}], X = [\text{“The”}, \text{“hippo”}, \text{“ate”}, \text{“my”}, \text{“homework”}])$$

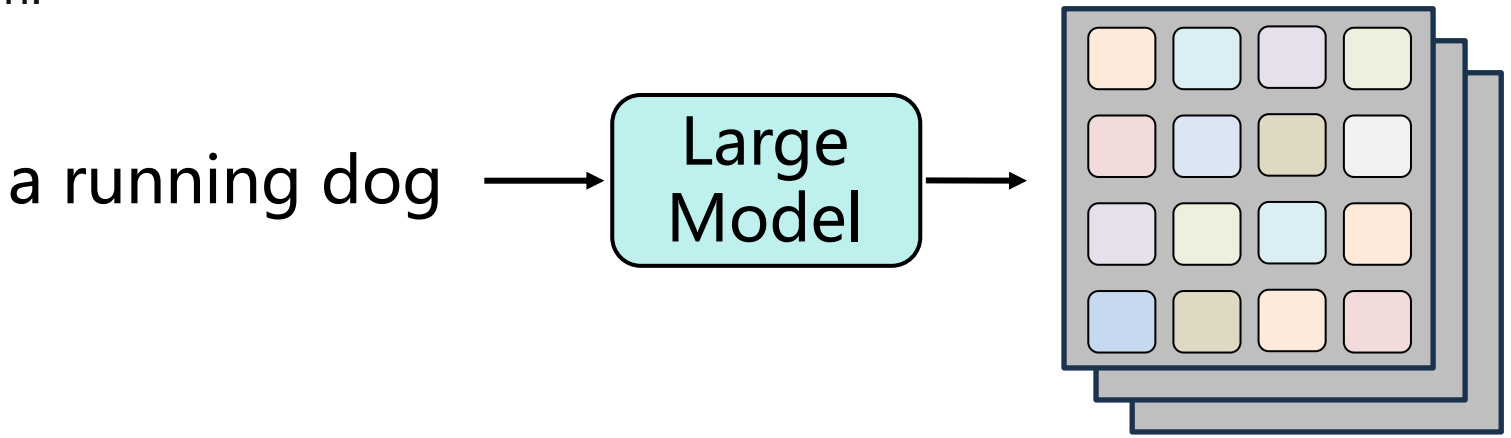
AI Large Model

How do large models generate images or videos?

Image Generation:



Video Generation:



AI Large Model

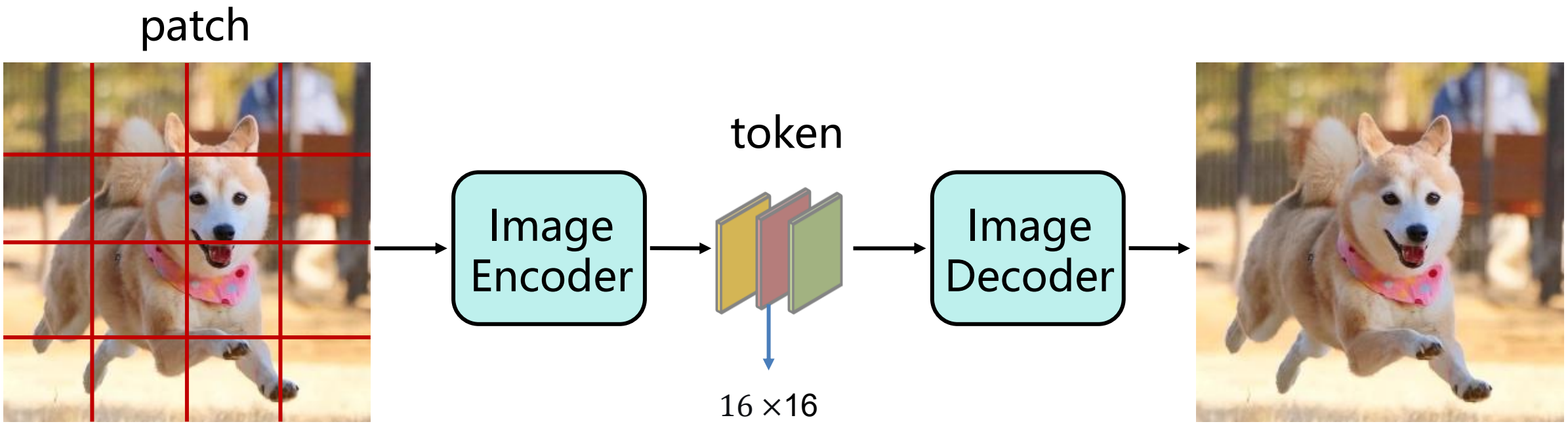
How do large models generate images or videos?

To generate a 1024×1024 image:

Tokens are regressively predicted by more than1,000,000 times

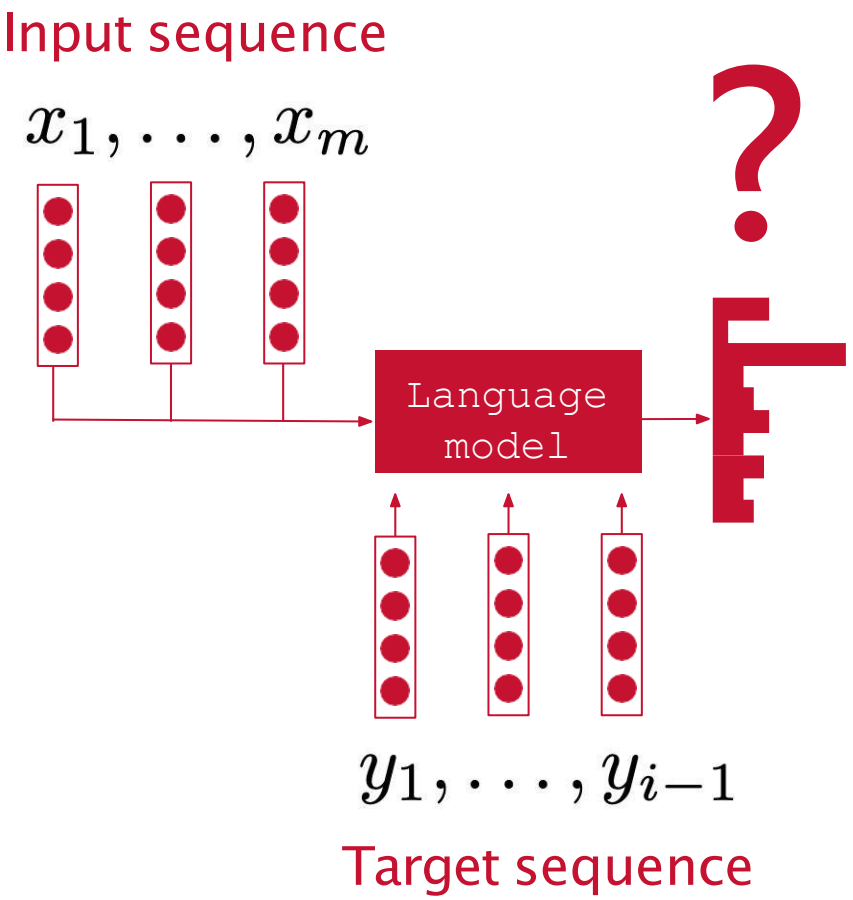
AI Large Model

How do large models generate images or videos?



AI Large Model

Summary



$P(Y_i)$

Or sometimes …

$P_\Theta(Y_i)$

Or sometimes …

$P(y_i \mid y_{1:i-1}, x_{1:m}, \Theta)$

Thank you!

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